



The top 10 chip categories

Ten chip types that decide your daily gadget experience

If you've ever wondered why two phones with "the same RAM" feel wildly different... or why one smart TV is buttery while another stutters... or why your router turns into a toaster the moment the whole family goes online – welcome to the actual story: your experience is the sum of multiple chips doing multiple jobs.

A modern gadget isn't powered by a chip. It's powered by a cast – each member with a specific role: compute, connectivity, power, display, storage, audio, sensors, and (increasingly) motor + energy control in vehicles and homes. The trick as a reader is to stop treating chip specs as trivia and start treating them as experience levers.

In this chapter, we'll map the 10 most common chip categories you're effectively "buying" every time you buy a phone, TV, laptop, router, earbuds, car, inverter, or even a power bank – and then we'll give you simple, step-by-step ways to *spot them, understand them, and buy smarter*.

Smartphone SoCs: the "everything chip"

A System-on-Chip (SoC) combines most of a computing system on one chip – typically CPU + GPU, plus a long list of controllers and accelerators (often ISP for camera, AI blocks, connectivity interfaces, and more).